

DSA for Data Science Fresher Interviews

Goal: interview-level DSA, not full software-engineering DSA. Focus on Python problem solving, patterns, time/space complexity, and clean explanations.

Recommended syllabus

Topic	What to learn	Priority
1. Arrays & Hashing	Arrays, frequency counting, sets/dictionaries, prefix sums, duplicates, two-sum patterns	HIGH
2. Strings	Character frequency, palindrome, anagram, substring, two pointers	HIGH
3. Two Pointers	Opposite pointers, same-direction pointers, sorted-array patterns	HIGH
4. Sliding Window	Fixed and variable windows, frequency maps	HIGH
5. Sorting	Python sorted/sort, custom key, merge/quick sort concepts, sorting-based patterns	HIGH
6. Binary Search	Classic binary search, first/last position, search on answer	HIGH
7. Stack	Parentheses, monotonic-stack basics, expression evaluation	MEDIUM
8. Queue / Deque	Basic queue, BFS idea, sliding-window maximum concept	MEDIUM
9. Linked List	Traversal, reverse, middle, cycle, fast/slow pointers	MEDIUM
10. Recursion	Base case, recursive thinking, simple tree recursion	MEDIUM
11. Trees	Binary tree traversal, depth, BST basics, level order	MEDIUM
12. Heap / Priority Queue	Min/max heap, top-K, kth largest/smallest	MEDIUM
13. Graph Basics	BFS/DFS concepts, visited set, connected components	LOW
14. Advanced DSA	Trie, advanced graphs, advanced DP, complex backtracking	SKIP FOR NOW

Focused LeetCode problem set

Target: ~55–60 problems. Solve Easy first; attempt Medium after understanding the pattern. You do not need to grind hundreds of problems for a Data Science fresher role.

Arrays & Hashing

- 1. Two Sum (Easy)
- 217. Contains Duplicate (Easy)
- 242. Valid Anagram (Easy)
- 49. Group Anagrams (Medium)
- 238. Product of Array Except Self (Medium)
- 347. Top K Frequent Elements (Medium)
- 128. Longest Consecutive Sequence (Medium)
- 53. Maximum Subarray (Easy)
- 121. Best Time to Buy and Sell Stock (Easy)
- 189. Rotate Array (Medium)

Strings / Two Pointers

- 125. Valid Palindrome (Easy)
- 344. Reverse String (Easy)
- 283. Move Zeroes (Easy)
- 167. Two Sum II (Medium)
- 15. 3Sum (Medium)
- 3. Longest Substring Without Repeating Characters (Medium)
- 424. Longest Repeating Character Replacement (Medium)

Sorting & Binary Search

- 88. Merge Sorted Array (Easy)
- 75. Sort Colors (Medium)
- 56. Merge Intervals (Medium)
- 704. Binary Search (Easy)
- 35. Search Insert Position (Easy)
- 34. Find First and Last Position of Element in Sorted Array (Medium)
- 33. Search in Rotated Sorted Array (Medium)
- 74. Search a 2D Matrix (Medium)

Stack / Queue

- 20. Valid Parentheses (Easy)
- 155. Min Stack (Medium)
- 496. Next Greater Element I (Easy)
- 739. Daily Temperatures (Medium)
- 150. Evaluate Reverse Polish Notation (Medium)
- 232. Implement Queue using Stacks (Easy)

Linked List

- 206. Reverse Linked List (Easy)
- 876. Middle of the Linked List (Easy)
- 141. Linked List Cycle (Easy)
- 21. Merge Two Sorted Lists (Easy)
- 19. Remove Nth Node From End of List (Medium)
- 234. Palindrome Linked List (Easy)

Trees / Heap

- 104. Maximum Depth of Binary Tree (Easy)
- 144. Binary Tree Preorder Traversal (Easy)
- 94. Binary Tree Inorder Traversal (Easy)
- 102. Binary Tree Level Order Traversal (Medium)
- 226. Invert Binary Tree (Easy)
- 98. Validate Binary Search Tree (Medium)
- 215. Kth Largest Element in an Array (Medium)
- 703. Kth Largest Element in a Stream (Easy)

Graph Basics

- 733. Flood Fill (Easy)
- 200. Number of Islands (Medium)
- 997. Find the Town Judge (Easy)

How to prepare each problem

1. Try independently for 20–30 minutes.
2. If stuck, study the pattern/hint rather than memorizing the code.
3. Re-code it without looking.
4. Explain the approach aloud: brute force → optimized approach → time complexity → space complexity.
5. Revisit failed problems after 3–7 days.

Priority for your Data Science preparation

DSA should support your interview preparation, not dominate it. After this core set, prioritize Python, SQL, statistics/probability, machine learning, deep learning/NLP, GenAI, and project interview questions.